

Abdullah Al Mahfuz

Department of CSE
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SUMMARY

A visionary and systems-driven robotics enthusiast with strong expertise in graphics design, video editing, and event logistics. Proven track record in leading branding efforts for university clubs, mentoring juniors in robotics prototyping and Arduino programming, and coordinating national-level competitions and workshops. Skilled in documentation design, process automation, and collaborative workflow optimization using tools like Google Sheets, Illustrator, and Apps Script. Passionate about community building, inclusive recruitment, and bridging technical precision with creative storytelling.

EDUCATION

Degree/Certificate	Institute	CGPA/GPA	Year
BSc (CSE)	United International University	2.39 (50 credit completed)	2023- Running
HSC (Science)	Ccer Model College	4.75 /5.00	2020
SSC (Science)	Zhinaiya High School	4.50 /5.00	2018

WORK EXPERIENCE

- Sub Executive, UIU Robotics Club**

United International University

Dhaka, Bangladesh
Jan 2024 – Mar 2025

 - Led club branding efforts by designing posters, banners, certificates, and social media content for events and outreach.
 - Assisted in organizing robotics workshops, competitions, and technical sessions.
 - Mentored juniors in robotics prototyping, Arduino programming, and competition preparation.
 - Helped build collaboration between UIU Robotics and external institutions.

VOLUNTEERING EXPERIENCE

- UIU CSE Project Show 242 & 243**

Department Project Showcase

Dhaka, Bangladesh
Summer & Fall 2024

 - Helped coordinate logistics and booth setup for student-led project exhibitions.
 - Assisted participants with technical demos and guided visitors through the showcase experience.
 - Supported session transitions to ensure smooth event flow.
 - Facilitated timely presentations and contributed to overall event organization and visitor engagement.
- UIU CSE Project Show 251**

Department Showcase

Dhaka, Bangladesh
2025

 - Led volunteer coordination and managed team schedules for the event.
 - Contributed to branding and promotional materials for outreach.
- UIU Robotics Rush Hours**

Robotics Competition

Dhaka, Bangladesh
2024

 - Volunteered in match coordination and supported event logistics.
 - Helped prepare event documentation, including team lists and the official rulebook.
 - Designed promotional materials such as posters, banners, and social media posts for event branding.
 - Helped coordinate logistics and booth setup for student-led project exhibitions.
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 - Assisted participants with technical demos and guided visitors through the showcase.

- **RESEARCH 101 Workshop**

Dhaka, Bangladesh

Academic Training Program

2025

- Volunteered during the event, assisting with logistics, participant onboarding, and speaker coordination.
- Designed the official poster, X-banner, and Facebook promotional post to support branding and outreach.
- Ensured visual consistency across print and digital materials using Adobe Photoshop and Illustrator.
- Helped distribute materials and maintained smooth flow during workshop sessions.

- **UIU CSE Fest**

Dhaka, Bangladesh

Departmental Festival

2025

- Volunteered across multiple segments including tech talks, competitions, and exhibitions.
- Managed crowd flow and helped maintain event schedule and coordination.

- **WROBD 2025 Regional Round**

Dhaka, Bangladesh

World Robot Olympiad Bangladesh

2025

- Assisted in organizing regional qualifiers for WROBD, supporting logistics and team coordination.
- Helped with field setup, scoring, and technical troubleshooting during matches.

- **WROBD 2025 National Round**

Dhaka, Bangladesh

World Robot Olympiad Bangladesh

2025

- Volunteered in the national-level robotics championship, supporting event execution and team services.
- Contributed to documentation, media coverage, and award ceremony coordination.

PRACTICAL EXPERIENCE

- **Robotics Enthusiast (Self-Driven)**

Dhaka, Bangladesh

Prototyping & Competitive Robotics

2024–2025

- Gained hands-on experience in designing, assembling, and testing robotics systems including Soccer Bots and Line Follower Robots (LFR).
- Applied control algorithms (e.g., PID) and integrated sensors, Arduino microcontrollers, and motor driver circuits for functional prototypes.
- Designed robot chassis and components in Fusion 360, ensuring durability and lightweight structure for competitions.
- Participated in multiple national robotics competitions, enhancing problem-solving, teamwork, and hardware troubleshooting skills under pressure.

- **Participant, Machine Learning Olympiad BD**

Dhaka, Bangladesh

National Competition

2024

- Developed and optimized ML solutions to predictive modeling problems using Python, NumPy, and Pandas under time constraints.
- Ranked among the top 50 nationally, demonstrating practical application of ML algorithms to real-world data-driven challenges.

- **IoT Workshop Attendee**

Dhaka, Bangladesh

Hands-on Training Program

2024

- Gained practical experience in IoT hardware integration, sensor interfacing, and microcontroller-based prototyping.
- Worked on small-scale IoT projects demonstrating connectivity between devices and cloud platforms.

AREA OF EXPERTISE

Graphics Design, Video Editing, 3D Designing, Machine Learning, Front-End Development, PCB Designing, Robotics (ROS), and Prototyping for Robotics and IoT.

RELEVANT COMPETITIONS

●Line Follower Robot (LFR) Competitions

- **KUET BitFest** | *Line Follower Robot (LFR) Segment* 2025
 - **UIU CSE Fest** | *Line Follower Robot (LFR) Segment* 2025
 - **UAP Techtron** | *Line Follower Robot (LFR) Segment* 2025
 - **BRACU Traction** | *Line Follower Robot (LFR) Segment* 2025
- * Competed in precision-based LFR challenges featuring complex track layouts with sharp turns, intersections, and speed zones.
 - * Tuned PID parameters for optimal line tracking performance across varying surface conditions.
 - * Achieved sub-second lap times and ranked among top performers in sensor calibration and control logic.
 - * Collaborated with teammates to debug real-time issues and refine motor control algorithms during competitions.

● Soccer Bot Competitions

- **EWU National RoboFest** | *Soccer Bot Segment* 2024
 - **UIU EEE Day** | *Soccer Bot Segment* 2024
 - **UIU Robotics Rush Hours** | *Soccer Bot Segment* 2024
 - **AUTOMECH** | *Soccer Bot Segment* 2025
- * Designed and deployed soccer-playing robots with wireless control and modular chassis for competitive matches.
 - * Focused on maneuverability, speed control, and strategic positioning during gameplay.
 - * Participated in multiple national-level events, refining robot performance through iterative testing and teamwork.
 - * Strengthened collaboration and troubleshooting skills under pressure in dynamic match environments.

● Machine Learning Olympiad

- **Machine Learning Olympiad BD** | *National Competition* 2024
- * Competed in the national-level Olympiad focusing on real-world applications of machine learning.
 - * Solved predictive modeling and AI-driven problem statements under time-constrained conditions.
 - * Ranked 43rd nationally for model accuracy and optimization.

ACHIEVEMENTS

- 5th place in Robotics Rush Hour 2024.
- Advanced to 2nd stage in EWU RoboFest 2024.
- Ranked 43rd nationally in Machine Learning Olympiad BD 2024.

PROBLEM SOLVING

I have solved a good amount of easy and medium problems from LeetCode and Hackerank.

PROJECTS

- Soccer Bot | *Robotics Project*

2024

- Designed and developed a soccer-playing robot using 37GB 555 gear motors (400 RPM) and motor drivers.
 - Integrated wireless remote control for maneuvering and tested performance in multiple robotics competitions.
 - Tools/Tech: Fusion 360 (design), Arduino (control).
- Line Follower Robot (LFR) | *Robotics Project*

2025

- Built a Line Follower Robot capable of following complex paths with high accuracy.
 - Implemented PID control for smoother and faster line tracking.
 - Tools/Tech: Arduino, IR sensors, motor driver circuits, Fusion 360 (chassis design).
 - Designed PCB for the robot using **EasyEDA**, integrating sensors and motor driver circuits efficiently.

LANGUAGE SKILLS (ENGLISH)

- Reading
- Writing
- Speaking

TECHNICAL EXPERTISE

- 3D Designing:** Fusion 360, Blender
- Machine Learning:** Problem solving with PyTorch, NumPy, Pandas
- Front-End Development:** HTML, CSS, JavaScript
- PCB Designing:** Circuit design, prototyping
- Robotics and IoT:** ROS, robot prototyping, line follower and soccer bots
- Graphics Design:** Adobe Photoshop, Illustrator